



Final project: student brief

Machine Learning with Python

Overview: For your final project, choose one of the following three options. Each project focuses on a different part of the course:

1. CNN classification and error analysis
2. Neural networks compared with a classical model
3. Unsupervised learning and clustering

You will receive a base notebook containing the imports, data loading, and initial preparation. Complete the remaining cells in order, following the comments and hints provided. You will also write several short interpretations in the marked markdown cells.

Submit your completed notebook with **all cells run and all outputs visible**. The data sets are downloaded when the notebooks run, so an internet connection is required the first time.

Option 1: Where does your CNN make its mistakes?

Scenario: A postal sorting centre and a bank both rely on software that reads handwritten numbers: postcodes on envelopes, account numbers on forms. A wrong reading is not free, since a misread postcode sends a letter to the wrong city and a misread account number can misdirect a payment. Your task is to build a model that reads handwritten digits and, just as importantly, to understand where it goes wrong, so its mistakes can be anticipated rather than found in production.

Train a convolutional neural network to recognise handwritten digits from the MNIST data set. After evaluating its performance, use a confusion matrix to identify the digit pairs it finds most difficult.

You will then make one controlled change to the network and investigate whether its accuracy and pattern of mistakes change. The aim is not simply to achieve a high score, but to understand what the model gets wrong and whether the modification produces meaningful improvement.

Central question: *Where does the CNN make its mistakes, and does one change to the model move them?*

Ethical considerations: the reliability of the model across different handwriting styles, and the consequences of a confident misreading in an automated pipeline.

Submit the completed notebook with both confusion matrices, the comparison of the two models, and your written responses.



Option 2: Does a neural network beat a classical model?

Scenario: A public agency wants to use survey data to flag which households are likely to be in a higher-income bracket, so that a benefit or outreach programme can be targeted. Two kinds of models are on the table: a simple, transparent one whose decisions can be explained to the public, and a more complex neural network that might, or might not, be more accurate. Your task is to build both, compare them fairly, and judge whether the added complexity earns its place.

Use the Adult Census Income data set to compare two approaches to the same classification problem: logistic regression and a small neural network.

The data contain both numerical and categorical features, so you will first construct an appropriate preprocessing pipeline. You will evaluate logistic regression using cross-validation, predict how you expect the neural network to perform, and then compare both models on the same test set.

Because the income classes are imbalanced, F1 will be your main comparison metric. The confusion matrices will help you examine whether the models make different kinds of errors.

Central question: *Does the neural network perform better and is its additional complexity worthwhile?*

Ethical considerations: fairness across the sex, race, and nationality groups in the data, the differing costs of false positives and false negatives, and the value of a transparent model when decisions affect people.

Submit the completed notebook with both confusion matrices, the model comparison, your prediction, and your final interpretation.



Option 3: How much structure can you find without labels?

Scenario: An agricultural co-op receives deliveries of mixed wheat grain and suspects it contains several distinct varieties, but the deliveries arrive unlabelled. Before investing in a labelling or sorting process, the co-op wants to know how many varieties are plausibly present and how cleanly they separate on a few simple measurements. Your task is to find that structure using clustering alone, then judge how much of it is real.

Explore the Seeds data set, which contains seven measurements from 210 wheat kernels. The true varieties will be kept aside while you investigate the structure of the measurements.

You will use PCA to visualise the data and apply K-Means clustering without consulting the labels. The elbow method and silhouette score will help you decide how many clusters the data supports. Only after completing the clustering will you reveal the true varieties and evaluate how closely they correspond to the groups you found.

Central question: *How much of the real class structure can clustering recover without using the labels?*

Ethical considerations: the risk of treating discovered clusters as real, fixed categories, and how the purity score and the choice of k should temper any claim.

Submit the completed notebook with the diagnostic charts, cross-tabulation, purity score, PCA plots, cluster profiles, and your written interpretations.